Self-distillation of context — cross-checkout report

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Self-distillation of context — cross-checkout comparison.

Method families compared (one row per run, best-effort artifacts):
  [layerwise]  158 runs
      Forward layerwise distillation (this checkout). Every block updated with uniform k-deep credit; behavioral readout teacher-sourced.
  [classical-kd]  5 runs
      Classical/naive KD baselines (../selfupdate_kd): full fine-tune and LoRA KL-distillation of the top layers.
  [multigpu]  18 runs
      Big-model layerwise on sharded GPUs (../selfupdate_multi): gpt-oss-120B, Qwen3.5-122B, Gemma-4, ALIA-40B.

Metric switch (owner directive, 2026-07-07): the tiny held-out general
cross-entropy (log-loss) 'destruction' canary was measured to be
noise-dominated and near-flat across training. It is replaced by two
standard, bounded, low-variance measurements, scored in a single
teacher-forced batched pass so only model LOADING costs time:
  DESTRUCTION (capability retained vs the epoch-0 teacher of the base):
    - ARC-Easy accuracy on a fixed cached subset (option log-likelihood)
    - WikiText-2 validation perplexity (quantization-style damage)
  RECALL (memorization of the original text; eval-only, never trained):
    - exact continuation + interior-word cloze (Machado verse)
    - next-sentence continuation + multi-word paragraph censor (Cervantes)

Coverage: 170 trained runs total; 129 with the new retention
battery, 147 with recitation CER. The index also includes 11
epoch-0 teacher/original baseline rows, where ARC retained = 1 and
WikiText log perplexity-ratio damage = 0. Runs still needing the battery
are listed on the coverage page; rerun scripts/retention_eval.py to fill them.

Generated 2026-07-07 14:40. Backbone: runs/retention_index.csv.
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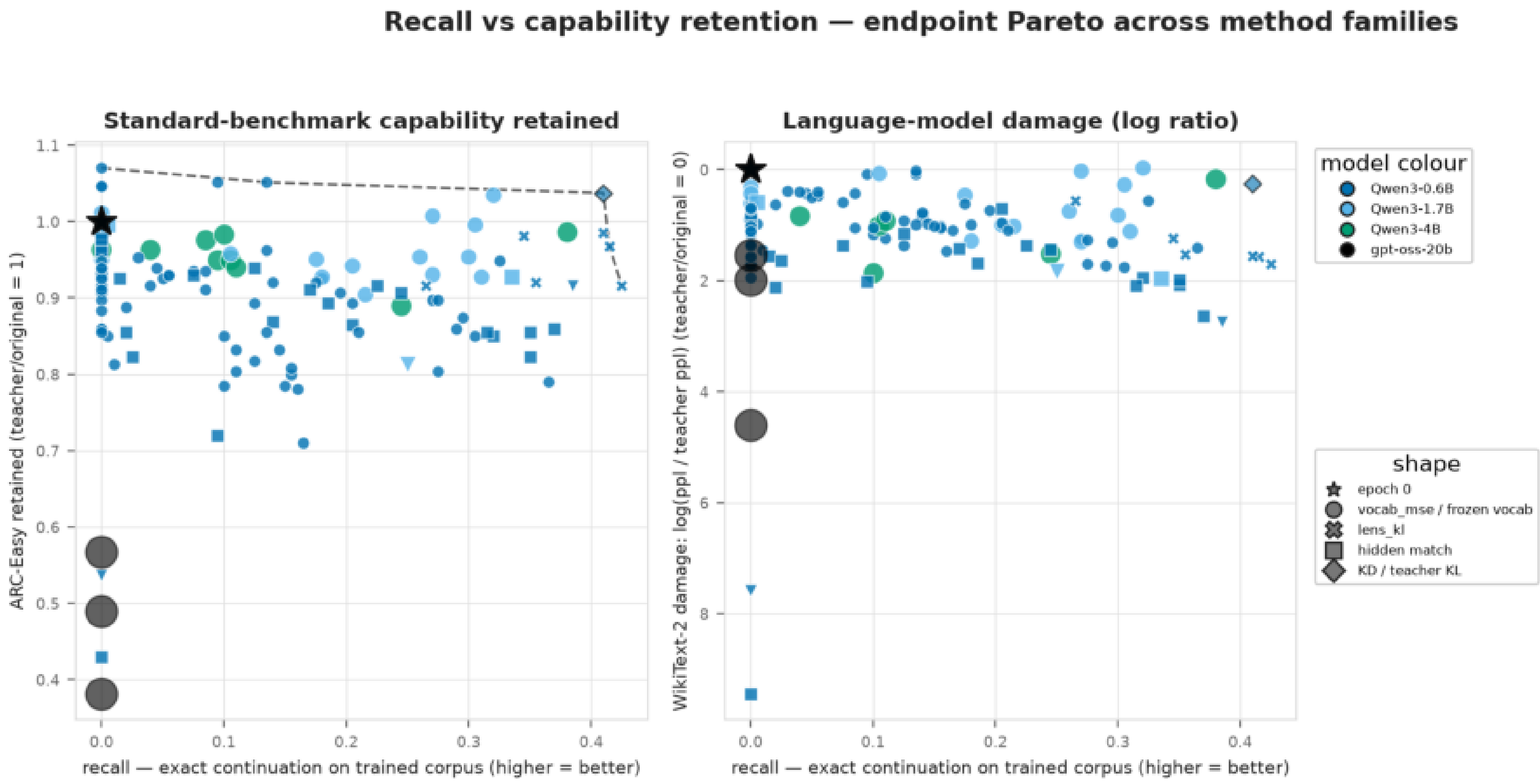
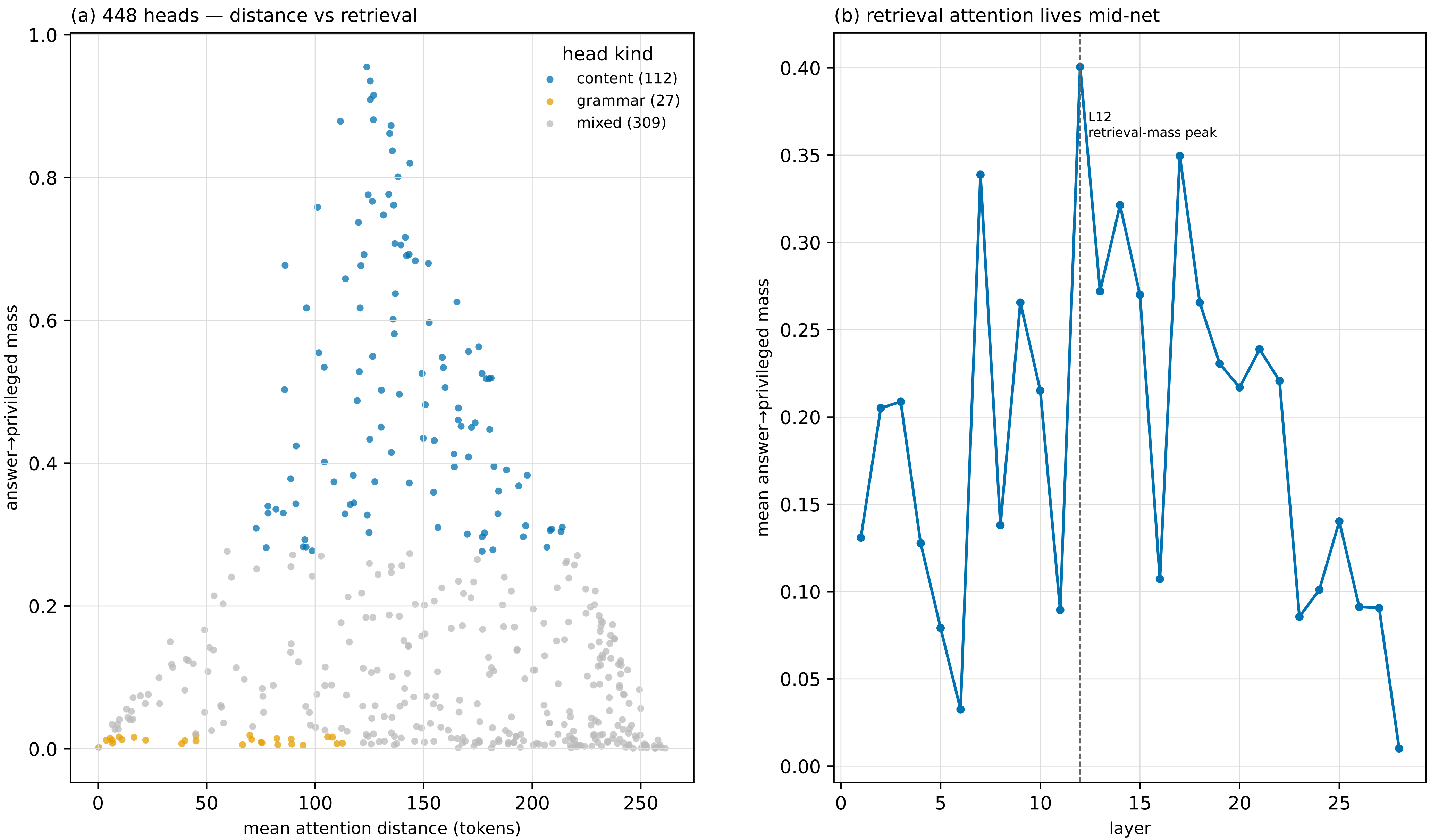
Recall vs capability retention (bidimensional)



140 rows with retention evidence, including epoch-0 teacher anchors · colour = model · shape = loss/lens/readout kind · marker size ~ model params

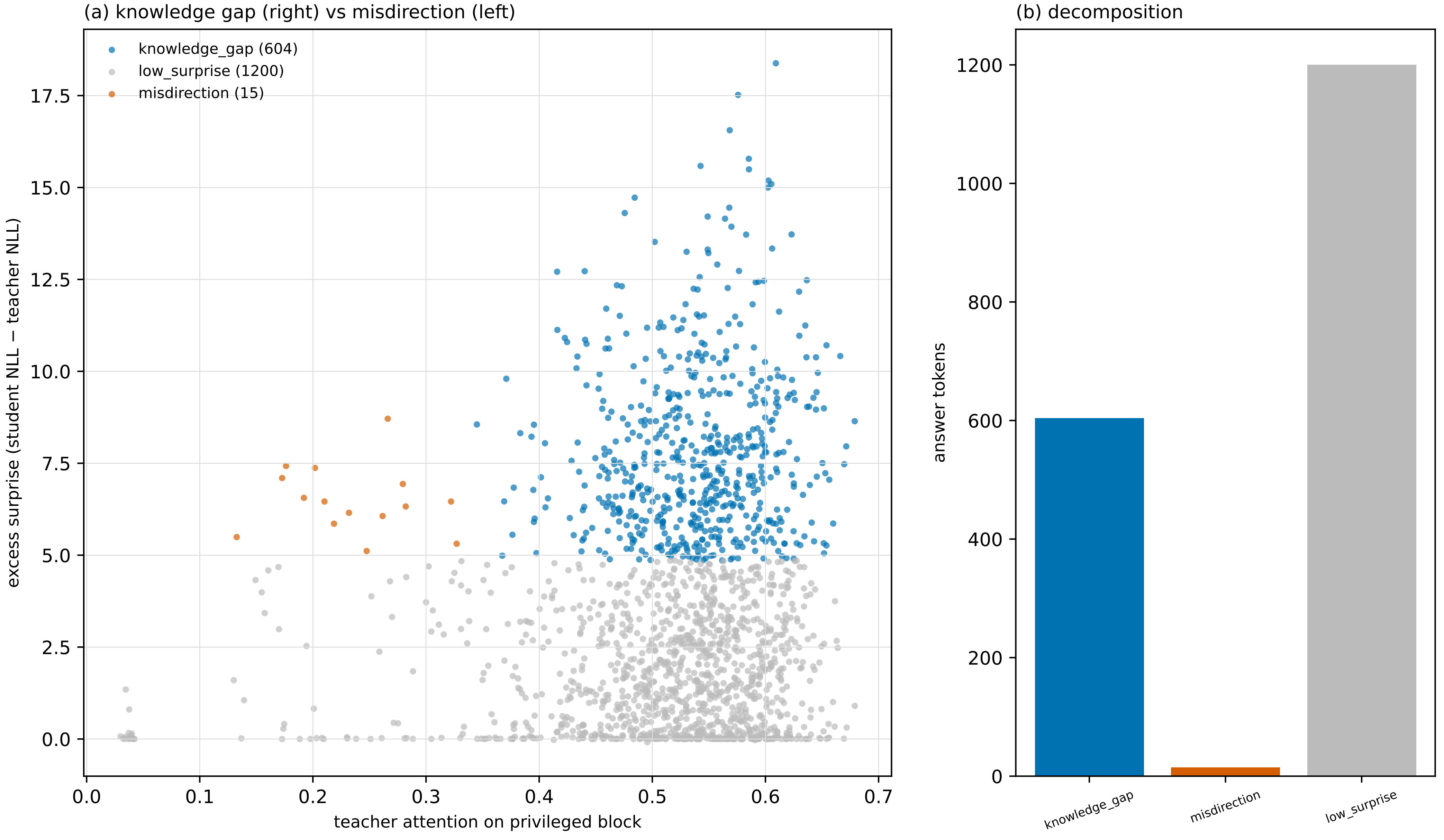
What to memorize — attention head taxonomy (0.6B)

Teacher view (privileged passage present), answer positions. A head that reaches FAR (high distance) and lands ON the privileged block (high answer→privileged mass) is a content/insight head — it marks what the context considered worth absorbing. Local, privileged-blind heads are grammar. This is the memorization target that motivates hidden-primary (mid-net) storage, and it names what the recall probes should test.



What to memorize — surprise decomposition (0.6B)

Surprise = student-view NLL minus teacher-view NLL on each answer token (base model, privileged block removed vs present). Footprint over content heads, structural/sink tokens masked. High surprise splits into knowledge_gap (604; teacher attends the privileged block) and misdirection (15; teacher attends shared in-context tokens). Knowledge gaps are the memorization target; misdirection is a routing fix.



Best retention row per method family and model

Recall: CER (lower better) and exact-match probes (higher better). Retention: ARC retained and WikiText log(ppl / teacher ppl) damage.

source	model	run	recall_cer	cont_mach	cloze_mach	censor_cerv	arc_acc	arc_retain	wiki_log_damage
layerwise	Qwen3-0.6B	clean_q_ch1_slide8_vocab_0p6b_	0.980	0.000	0.009	0.886	0.573	1.070	0.685
layerwise	Qwen3-1.7B	dev	—	0.320	0.936	0.102	0.677	1.034	-0.029
layerwise	Qwen3-4B	lw_l_tailonly_4b_rag	0.013	0.380	0.982	0.136	0.738	0.987	0.160
classical-kd	Qwen3-0.6B	kd_full_0p6b_rag	0.863	0.410	0.991	0.034	0.555	1.037	0.258
multigpu	gpt-oss-20b	mlt_gptoss20b_slide2_tf_lora	—	0.000	0.000	0.011	0.420	0.568	1.537

Retention coverage — remaining work

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Runs WITHOUT the new retention battery (need scripts/retention_eval.py):

[layerwise] 24 missing
   clean_combined_slide8_vocab_14b_lora  clean_combined_slide8_vocab_8b_lora  clean_machado_slide8_vocab_14b_lora
   clean_machado_slide8_vocab_8b_lora  clean_machado_slide8_vocab_alia40b_lora  clean_q_ch1_slide2_vocab_14b_lora
   clean_q_ch1_slide4_vocab_14b_lora  clean_q_ch1_slide8_lenskl_14b_lora  clean_q_ch1_slide8_nmse_14b_lora
   clean_q_ch1_slide8_vocab_14b_lora  clean_q_ch1_slide8_vocab_8b_lora  clean_q_ch1_slide8_vocab_alia40b_lora
   lw_j_llama8b_rag  lw_j_mistral7b_rag  lw_j_phi4mini_rag
   lw_j_qwen8b_rag  lw_k_gptoss_rag  lw_l_final_14b_rag
   lw_l_final_8b_av2_rag  lw_l_final_8b_rag  lw_p_ft4b_rag
   lw_tc_lora_14b_rag  smoke_bucketed_mb2_gpu3  smoke_item_mb2_gpu3

[classical-kd] 3 missing
   kd_lora_kl_hi_e40_v3_qwen36_27b_rag  kd_lora_kl_hi_e40_v3_qwen3_30ba3b_inst2507_rag  kd_lora_kl_hi_e60_v3_14b_rag

[multigpu] 14 missing
   mlt_alia40b_slide2_lora  mlt_gemma4_26ba4b_slide2_lora  mlt_gemma4_26ba4b_slide2_ra_lora
   mlt_gemma4_26ba4b_slide2_tf_lora  mlt_gemma4_31b_k1local_lora  mlt_gemma4_31b_slide2_lora
   mlt_gemma4_31b_slide3_lora  mlt_gptoss120b_auto4_slide2_tf_lora  mlt_gptoss120b_pp2_k1local_lora
   mlt_gptoss120b_pp2_slide2_lora  mlt_gptoss120b_pp2_slide2_tf_lora  mlt_gptoss120b_pp2_slide3_lora
   mlt_qwen35_122b_auto4_slide2_lora  mlt_qwen35_122b_gptq_auto2_slide2_lora
```